Kraemer / Ladera / Jeffrey Trail Scrimmage - Repaired practice edition - Answer key

Division B | Thermodynamics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

reasoning: Q=+500 J, Wby=+200 J.
delta U=500-200=300 J.
won=-200 J, giving q+won=300 J. (3 points).

conclusion: Both sign conventions give the same energy change. (3 points).